gpeyt1-37 · Overturning FS

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 12 ft tall plain-concrete gravity wall (γ_c = 150 pcf) has a vertical back face and a battered front face. Crest width 2 ft, base width 7 ft (toe at the front). Backfill is level, dry sand: γ = 120 pcf, φ = 30°, c = 0, no wall friction. Passive resistance in front is neglected. The foundation is dense gravel with δ = 28°.

Most nearly, the factor of safety against overturning is:

A. 2.12
B. 0.96
C. 3.18
D. 1.59

Given equations / data

> $K_a = \tan^2(45 - \varphi/2)$; $P_a = \frac{1}{2} K_a \gamma H^2$ at H/3; FSOT = ΣMresisting/ΣMoverturning about the toe

Worked solution

1. $K_a = 1/3$; $P_a = \frac{1}{2}(1/3)(120)(12)^2 = 2{,}880$ lb/ft at 4.0 ft
2. Rectangle (back): W1 = 2×12×150 = 3,600 lb at 6.00 ft from toe
3. Triangle (front): W2 = ½×5×12×150 = 4,500 lb at 3.33 ft
4. MR = 36,600; MO = 2,880×4.0 = 11,520 ft-lb/ft
5. FSOT = 3.18

Final answer

3.18

Why the other values are wrong

- **2.12** — applied Pa at H/2
- **0.96** — took moments about the heel-side distances (wrong lever arms)
- **1.59** — forgot the ½ in Pa

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.18 / wall stability figures (moments about toe), p. 209

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=P-W_f8IyVos (no creator values copied).

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